

APPENDIX J

TIME MEASUREMENT SYSTEMS

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International Atomic Time, TAI, which is provided by atomic clocks, is the basis for the two time systems used in spacecraft time measurements. *Ephemeris Time, ET*, which is used in the preparation of ephemerides, is a uniform or "smoothly flowing" time and is related to *TAI* by

$$ET = TAI + 32.18 \text{ sec}$$

In contrast to *ET*, *Coordinated Universal Time, UTC*, uses the *TAI* second as the fundamental unit, but introduces 1-second steps occasionally to make *UTC* follow the nonuniform rotation of the Earth. *UTC* is necessary for terrestrial navigation and surveying for which the rotational position of the Earth at a given instant is critical. It is this time which is broadcast internationally and is used for tagging spacecraft data and for all civil timekeeping. Finally, *sidereal time* is a direct measure of the rotational orientation of the Earth relative to the "fixed" stars and, therefore, is used to estimate the position of a spacecraft relative to points on the Earth's surface. The characteristics of the various time systems are summarized in Table J-1.

Any periodic phenomenon may be used as a measure of time. The motion of the Earth, the Moon, and the Sun relative to the fixed stars has traditionally been used for this purpose. However, the need for increasingly accurate time measurements has resulted in the development of several alternative time systems. The requirement for high accuracy comes from the cumulative effect of timing errors. An accuracy of 1 sec/day (1 part in 10^5) would appear satisfactory for most scientific or technical purposes. However, an error of this magnitude in an ephemeris of the Earth causes an error of 10,800 km in the position of the Earth

Table J-1. Time Systems

KIND OF TIME	DEFINED BY	FUNDAMENTAL UNIT	REGULARITY	USE
SIDEREAL	EARTH'S ROTATION RELATIVE TO STARS	SIDEREAL DAY, 1 ROTATION OF EARTH	IRREGULAR	ASTRONOMICAL OBSERVATIONS; DETERMINING UT AND ROTATIONAL ORIENTATION OF EARTH
SOLAR:				
APPARENT	EARTH'S ROTATION RELATIVE TO TRUE SUN	SUCCESSIVE TRANSITS OF SUN	IRREGULAR AND ANNUAL VARIATIONS	SUNDIALS
MEAN	EARTH'S ROTATION RELATIVE TO FICTITIOUS MEAN SUN	MEAN SOLAR DAY	IRREGULAR	—
UNIVERSAL				
UT0	OBSERVED UT	MEAN SOLAR DAY	IRREGULAR	STUDY OF EARTH'S WANDERING POLE
UT1	CORRECTED UT0	MEAN SOLAR DAY	IRREGULAR	SHOWS SEASONAL VARIATION OF EARTH'S ROTATION
UT2	CORRECTED UT1	MEAN SOLAR DAY	IRREGULAR	BASIC ROTATION OF EARTH
UTC = GMT + Z	ATOMIC SECOND AND LEAP SECONDS TO APPROXIMATE UT1	MEAN SOLAR DAY	UNIFORM EXCEPT FOR LEAP SECONDS	CIVIL TIMEKEEPING; TERRESTRIAL NAVIGATION AND SURVEYING; BROADCAST TIME SIGNALS
EPHEMERIS, ET	FRACTION OF TROPICAL YEAR 1900	EPHEMERIS SECOND	UNIFORM	EPHEMERIDES
ATOMIC, TAI	FREQUENCY OF 133 Cs RADIATION	ATOMIC SECOND = EPHEMERIS SECOND	UNIFORM	BASIS OF ET AND UTC

after only 1 year, several orders of magnitude worse than what is acceptable for many unsophisticated measurements. Thus, the generation of accurate ephemerides requires a precise time measurement system.

The diurnal motion of celestial objects is the most obvious timekeeper. Until the Middle Ages "seasonal hours," one-twelfth of daylight or nighttime periods, was used. Of course, this unit varies both with the season and with the observer's latitude. A more uniform unit of time is the *apparent solar day*, defined as the interval between two successive passages of the Sun across the observer's meridian. As discussed below, this interval varies throughout the year due to variations in the Earth's orbital speed and the inclination of the ecliptic. The Earth's orbital motion does not affect the *sidereal day*, the interval between two successive meridian passages of a fixed star. However, irregularities in the rotation of the Earth cause both periodic and secular variations in the lengths of the sidereal and solar days.

The annual motions of celestial objects provide a measurement of time which is independent of the irregular variations in the rotation of the Earth. The *tropical year*, upon which our calendar is based, is defined as the interval of time from one vernal equinox to the next. The *ephemeris second* is defined as $1/31556925.9747$ of the tropical year for 1900. Because of the precession of the equinoxes (Section 2.2.2), the tropical year is about 20 minutes shorter than the orbital period of the Earth relative to the fixed stars. This latter period is known as the *sidereal year*. Due to secular variations in the orbit and rate of precession of the Earth, the lengths of both types of year (in units of s_e , the ephemeris second) vary to first order according to the relations:

$$\text{Tropical year} = 31556925.9747 - .530T$$

$$\text{Sidereal year} = 31558149.540 + .010T$$

where T is the time in units of Julian Centuries of 36525 days from 1900.0 [Newcomb, 1898].

The first satisfactory alternative to celestial observations for the measurement of time was the *pendulum clock*. The period of a pendulum is a function of the effective acceleration of gravity, which varies with geography and the position of the Sun and the Moon. The resonance frequency of quartz crystals has recently been employed in clocks; this frequency depends on the dimensions and cut of the crystal and its age, temperature, and ambient pressure. Atomic clocks are based on the frequency of microwave emission from certain atoms. An accuracy of 10^{-14} (fractional standard deviation) may be achieved with atomic clocks; corresponding accuracies for quartz and pendulum clocks are 2×10^{-13} and 10^{-6} , respectively. For an extended discussion of time systems, see Woolard and Clemence [1966], the *Explanatory Supplement to the Astronomical Ephemeris and the American Ephemeris and Nautical Almanac* [H.M. Nautical Almanac Office, 1961] and Müller and Jappel [1977].

Solar Time. The *celestial meridian* is the great circle passing through the celestial poles and the observer's zenith. As shown in Fig. J-1, the *hour angle*, HA , is the azimuthal orientation of an object measured westward from the celestial meridian. As the Earth rotates eastward, a celestial object appears to move westward and its HA increases with time. It takes 24 hours for an object to move completely around the celestial sphere or 1 hour to move 15 deg in HA ; thus, 1 deg of HA corresponds to 4 minutes of time.

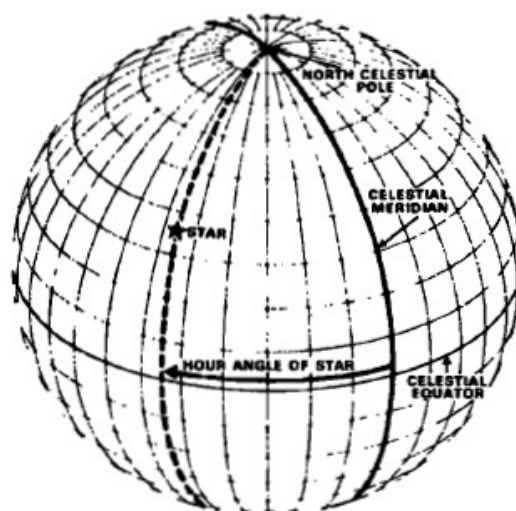


Fig. J-1. Definition of Hour Angle

The *apparent solar time* is equal to the local *HA* of the Sun, expressed in hours, plus 12 hours. Apparent solar time can be measured with a simple sundial constructed by driving a long nail perpendicularly through a flat piece of wood. If the nail is then pointed toward the celestial pole, the plane of the wood is parallel to the equatorial plane, and the shadow of the nail cast by the Sun onto the wood is a measure of the *HA*.

Due to the Earth's orbital motion, the Sun appears to move eastward along the ecliptic throughout the year. Because the Earth travels in an elliptical orbit, it moves faster when near the Sun and slower when it is more distant; therefore, the length of the solar day varies. Even if the Earth were in a circular orbit with a constant speed, the azimuthal component of the Sun's motion (parallel to the celestial equator) would vary due to the inclination of the ecliptic relative to the equator. To illustrate this, consider a satellite in a nearly polar orbit, as shown in Fig. J-2. The satellite changes azimuth slowly while near the equator and rapidly while near the poles. Although the variation in the length of the day due to the



Fig. J-2. Variation in Azimuthal Rate for a Satellite Moving Uniformly in its Orbit. $A_1, A_2, \dots, A_5$ are azimuthal projections of the orbital points 1, 2, ..., 5 and are equally spaced in time.

eccentricity and inclination of the Earth's orbit is small, the cumulative variation reaches a maximum of 16 minutes in November.

To provide more uniform time than the real Sun, a fictitious *mean Sun*, which moves along the equator at a constant rate equal to the average annual rate of the Sun, has been introduced. *Mean solar time* is defined by the *HA* of the mean Sun. The difference between the mean and apparent solar times is called the *equation of time*.

Standard Time. Mean solar time is impractical for communication and transportation because it varies continuously with longitude. Therefore, the world has been divided into 24 time zones of approximately 15 deg each. Normally, these zones are centered on standard meridians which are multiples of 15 deg in longitude. The uniform time throughout each zone is referred to as *Standard Time*, and usually differs by an integral number of hours from the mean solar time at 0 deg longitude, or *Universal Time*, as discussed below. Table J-2 lists the standard meridians for time zones in the continental United States. The apparent solar time is converted to Standard Time by adding the equation of time for the date and subtracting the algebraic difference (expressed in units of time) between the observer's longitude and the standard meridian.

Table J-2. Standard Time

TIME ZONE	STANDARD MERIDIAN (DEG, EAST LONG.)	UT MINUS STANDARD TIME (HOURS)	UT MINUS DAYLIGHT TIME (HOURS)
EASTERN	285	5	4
CENTRAL	270	6	5
MOUNTAIN	255	7	6
PACIFIC	240	8	7

Greenwich Mean Time, Universal Time. The 0-deg longitude line is referred to as the *Greenwich meridian* because it is defined by the former site of the Royal Greenwich Observatory. *Greenwich Mean Time*, *GMT*, is the mean solar time at 0 deg longitude; that is, *GMT* is the *HA* of the mean Sun observed at Greenwich (called the *GHA*) in hours plus 12 hours, modulo 24. *Greenwich Mean Time* is also called *Universal Time*, *UT*, and, in spaceflight operations, *Zulu*, or *Z*.

Uncorrected *UT* or *UT0* (read "UT Zero") is found from observations of stars, as explained in the discussion of sidereal time below. *UT0* time as determined by different observatories is not the same, however, due to changes in the longitudes of the observatories caused by the wandering of the geographic pole. Therefore, *UT0* is corrected for this effect to give *UT1*, which is then a measure of the actual angular rotation of the Earth. The Earth's rotation is subject to periodic seasonal variations, apparently caused by changes in, for example, the amount of ice in the polar regions. When *UT1* is corrected by periodic terms representing these seasonal effects, the result is *UT2*. Even *UT2* is not a uniform measure of time. Evidence from ancient eclipse records and other sources shows that the Earth's rate of rotation is slowing; also, unpredictable irregularities in the rotation rate are observed.

Before 1972, the broadcast time signals were kept within 0.1 sec of *UT2*. Since January 1, 1972, however, time services have broadcast *Coordinated Universal*

Time, UTC. A second of *UTC* is equal to a second of International Atomic Time, but *UTC* is kept within 0.90 sec of *UT1* by the introduction of 1-sec steps, usually at the end of June and December.

Ephemeris Time. The irregularities in the Earth's rotation cannot be predicted; however, gravitational theories have been formulated for the orbital motions of the Earth, the Moon, and the planets. In particular, Simon Newcomb's *Tables of the Sun*, [1898], published at the end of the 19th Century, gives the position of the Sun for regular time intervals. These intervals define a uniform time called *Ephemeris Time, ET*. In theory, Ephemeris Time is determined from observations of the Sun. In practice, observations of the Moon are used because the Sun moves slowly and its position is difficult to observe. One method is to record the *UT* of a lunar occultation of a star; the tabulated value of *ET* for the observed lunar position, corrected for effects such as parallax, is noted and the difference

$$\Delta T = ET - UT$$

is determined. A table of approximate ΔT values, both in the past and extrapolated into the future, is provided in *The American Ephemeris and Nautical Almanac* [U.S. Naval Observatory, 1973]. Ephemeris time at any instant is given by

$$ET = \Delta T + UT$$

International Atomic Time. The cesium nuclide, ^{133}Cs , has a single outer electron with a spin vector that can be either parallel or antiparallel to that of the nucleus. The flip from one orientation to the other, a hyperfine transition, is accompanied by the absorption or emission of microwave radiation of a given frequency. In an atomic clock, the number of these transitions is maximized in a resonator by the introduction of microwave radiation from an oscillator tuned to the same frequency. The cycles of the oscillator are counted to give a unit of time. In 1967, the 13th General Conference on Weights and Measures established the *Système Internationale (SI) second* as the duration of 9 192 631 770 periods of the radiation from the above transition in ^{133}Cs . This unit is the basis of *International Atomic Time, TAI*, and was chosen to make the SI second equal to the ephemeris second. The reference epoch for *TAI* is January 1, 1958, when $0^{\text{h}}0^{\text{m}}0^{\text{s}}$ *TAI* equaled $0^{\text{h}}0^{\text{m}}0^{\text{s}}$ *UT2*. For most purposes, ephemeris time may be considered to be equal to *TAI* plus 32.18 sec, the value of ΔT for January 1, 1958.

Sidereal Time. *Sidereal time, ST*, is based on the rotation of the Earth relative to the stars and is defined as the *HA* of the vernal equinox, Υ . The *local sidereal time, LST*, is defined as the local *HA* of Υ , $LHA \Upsilon$, and the *sidereal time at Greenwich, GST*, is defined as the Greenwich *HA* of Υ , $GHA \Upsilon$. Sidereal time may also be determined from the *HA* and *right ascension, RA*, of any star. The *RA* of a star is the azimuthal component of the star's position measured eastward from Υ (see Section 2.2.2). From Fig. J-3 we see that

$$LST = LHA \Upsilon = LHA^* + RA^*, \text{ modulo } 24 \quad (\text{J-1})$$

where LHA^* and RA^* are the *HA* and *RA* (both converted to time) of the star. In the example in Fig. J-3, LHA^* is 135 deg or 9 hours, RA^* is 90 deg or 6 hours, and the *LST* is 15 hours. Similarly,

$$GST = GHA \Upsilon = GHA^* + RA^*, \text{ modulo } 24 \quad (\text{J-2})$$

Day number of July 4.125 (= 3 ^h UT July 4), 1976	186.125
+ <i>JD</i> for January 0.0, 1976	+ 2 442 777.500
= <i>JD</i> for July 4.125, 1976	= 2 442 963.625
- <i>JD</i> for January 0.5, 1900	- 2 415 020.000
= <i>T</i> in days	27 943.625
÷ 36,525 = <i>T</i> in Julian centuries	0.765054757
8640184.542 <i>T</i> + 0.0929 <i>T</i> ²	6 610 214.340 sec
	= 76 ^d 12 ^h 10 ^m 14 ^s .340
+ first term Eq. (J-4)	6 38 45.836
+ <i>UT</i>	3 0 0.000
<i>GST</i>	21 ^h 49 ^m 0 ^s .176

para cálculo do GST, veja a ref. 'Escobal-parcial'.

Due primarily to the varying distances of the Sun and the Moon, a small amplitude oscillation, known as *astronomical nutation*, is superimposed on the precession of the equinoxes. Sidereal time corrected for this effect is called *mean sidereal time*. *GST* in Eqs. (J-4) and (J-5) is mean sidereal time. The maximum difference between mean and apparent sidereal time is only about 1 sec.

Table J-3. Julian Date at the Beginning of Each Year From 1975 to 2000

YEAR	JD FOR JAN 0.0 UT	YEAR	JD FOR JAN 0.0 UT	YEAR	JD FOR JAN 0.0 UT
	2 400 000+		2 400 000+		2 400 000+
1975	42 412.5	1984	45 699.5	1993	48 987.5
1976	42 777.5	1985	46 065.5	1994	49 352.5
1977	43 143.5	1986	46 430.5	1995	49 717.5
1978	43 508.5	1987	46 795.5	1996	50 082.5
1979	43 873.5	1988	47 160.5	1997	50 448.5
1980	44 238.5	1989	47 526.5	1998	50 813.5
1981	44 604.5	1990	47 891.5	1999	51 178.5
1982	44 969.5	1991	48 256.5	2000	51 543.5
1983	45 334.5	1992	48 621.5		

Because of the orbital motion of the Earth, a solar day is longer than a sidereal day. As illustrated in Fig. J-4, the fixed stars are sufficiently far away that lines connecting one of them to the Earth are essentially parallel. Because the Earth's orbital period is approximately 360 days, angle *A* is approximately 1 deg. A *sidereal day* is defined as one complete rotation of the Earth, 360 deg, relative to the stars. The Earth has to rotate 360 + *A* deg to complete a solar day. The ratio of the mean solar day to the mean sidereal day is 1.00273 79093; the mean sidereal day equals 23 hours, 56 min, 4.09054 sec of mean solar time and the mean solar day equals 24 hours, 3 min, 55.55536 sec of mean sidereal time [U.S. Naval Observatory, 1973]. Note that the "76 days" in the above example indicates the excess number of sidereal days, one for each year, that had occurred since the beginning of the century.

Sidereal time and mean solar time are affected proportionally by variations in the Earth's rotation. Although the irregular fluctuations in the Earth's rotation cannot be predicted, the general deceleration can be seen in Fig. J-5. The lengths of

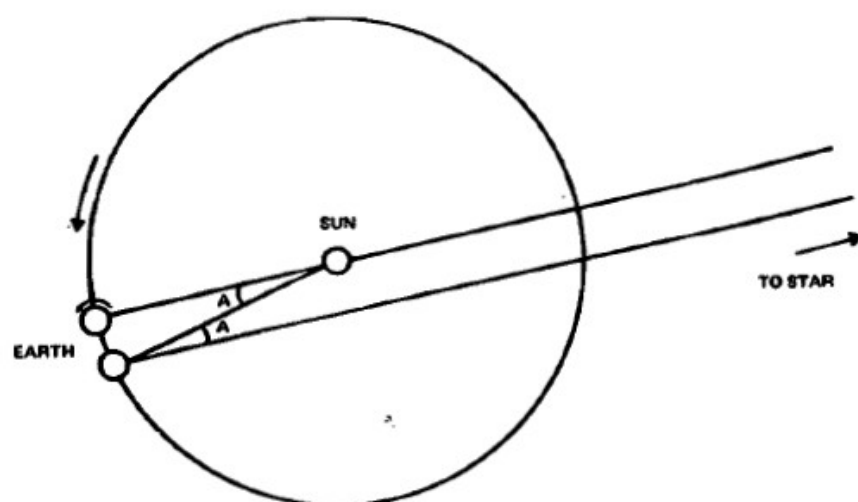


Fig. J-4. Orbital Motion of the Earth for 1 Day (Exaggerated) as Viewed From North Ecliptic Pole

the two types of day in s_e are given approximately by:

$$\text{Sidereal day} = 86164.09055 + 0.0015 T$$

$$\text{Mean solar day} = 86400 + 0.0015 T$$

where T is in Julian centuries from 1900.0 [Allen, 1973]. These terms are the dashed line in Fig. J-5.

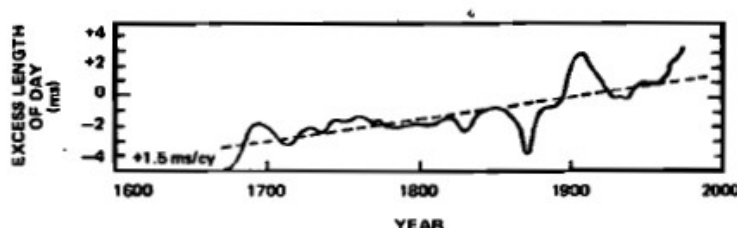


Fig. J-5. Excess Length of the Day Compared With the Day Near 1900. Note the very irregular fluctuations about the mean slope of 1.5 ms/century [Morrison, 1973].

Using Sidereal Time to Compute the Longitude of the Subsatellite Point. To determine the direction of geographic points on the Earth as seen from a spacecraft, it is necessary to know both the spacecraft ephemeris and the longitude of the subsatellite point. For any UT , Eq. (J-4) can be used to determine the Greenwich sidereal time, GST , which in turn can be used to determine the East longitude, EL_{spc} , of the subsatellite point for any satellite for which the right ascension of its position in geocentric coordinates is known. From Eq. (J-3), we have

$$EL_{spc} = RA_{spc} - GST \text{ (in degrees)}$$

where RA_{spc} is the right ascension in degrees of the spacecraft at time GST . Because UTC is accurate to about 1 sec, the accuracy of the resulting longitude will be about 0.005 deg if the spacecraft ephemeris is known precisely.

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